

Jaiveer Bassi

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Experience

Stanford University

Stanford SSG Fellow

Sep 2026 – Present

Stanford, CA

- Conduct mentor-guided independent research by developing a core curiosity question, evaluating primary scientific literature, and formulating testable hypotheses.
- Prepare research findings for a formal presentation to Stanford mentors and peers.

Physical Intelligence

AI Robot Operator

May 2026 – Present

San Francisco, CA

- Generate high-quality demonstration data for general-purpose robotics foundation models by teleoperating robotic arms through manipulation, sorting, and assembly tasks.
- Evaluate policy rollouts against task criteria, identify failure modes and edge cases, annotate executions, and enforce data-quality thresholds used in model development.
- Triage recurring hardware-software faults across the robotics stack, trace service logs to isolate root causes, and document reproducible findings for engineering teams.

Handshake AI

Contracted AI Response Evaluator

Apr 2026 – Jun 2026

Remote

- Evaluated LLM-generated code and technical responses across Python, C#/.NET, AI/ML, and software engineering for correctness, instruction following, and technical clarity.

Castline Studio

Operator & Developer – Resin 3D Printing, Next.js, TypeScript

Jan 2026 – Apr 2026

Vallejo, CA

- Operate resin 3D-printing production for an e-commerce business generating \$2K+/month across 100+ product listings, including calibration, curing, post-processing, and tolerance control.
- Built an order-management platform with client-side STL parsing for automated geometry, volume, dimension, and print-cost calculations.

Kaiser Permanente Regional Laboratory

Software Developer

Oct 2023 – Jun 2024

Berkeley, CA

- Developed a full-stack application using React, Express, RESTful APIs, and SQL for uploading, processing, and reviewing clinical laboratory instrument data across multiple departments.
- Added validation and error logging to a 50+ column instrument-data workflow; diagnosed faulty recordings and documented recurring failure patterns to improve pipeline reliability.

Selected Projects

π 0 on a Budget | OpenPI, LoRA, LeRobot, JAX, WSL2

- Built a synchronized data pipeline for recording 30 FPS Logitech Brio video and joint commands from a custom 4-DOF teleoperation arm, with conversion into a LeRobot-compatible dataset.
- Implemented an RTX 4070 workflow for π 0-FAST fine-tuning, memory profiling, policy serving, open- and closed-loop evaluation, and a real-time dashboard of live Arduino telemetry and RTX-predicted action chunks.

UR5e Fault Injection & Diagnostics Harness | RTDE, URSim, Fault Injection

- Built a control and fault-characterization harness that drives simulated robot motion and logs synchronized joint telemetry while testing connection failures, stale handles, hangs, and native crashes.

OpenPI Language-Steerability Demo | OpenPI, π 0.5, LIBERO, BDDL

- Ran the public π 0.5 LIBERO checkpoint in one simulated scene under different instructions, scored by the simulator's own goal checks: trained and paraphrased prompts succeeded 10/10, novel object recombinations as low as 1/10.

GELLO-Style Leader-Follower Teleoperation Rig | Mechanical Assembly, Servo Control, Calibration

- Built a four-DOF leader-follower mechanism for approximately \$30 by assembling linkages, wiring potentiometer-based joint encoders, and calibrating direct joint-space control without inverse kinematics.

Scribbot Desktop Drawing Arm | OpenSCAD, 3D Printing, Arduino, PCA9685

- Designed in OpenSCAD and built a 3D-printed four-servo drawing arm run by an Arduino Nano and PCA9685.

CAN Bus Fault-Injection & Diagnostics Rig | *SocketCAN, MCP2515, Multimeter Diagnostics*

- Assembled a three-node CAN network with Arduino/MCP2515 nodes, a USB-CAN interface, and a custom wiring harness, then induced bus-off, arbitration, connector, wiring, and termination faults.
- Mapped physical failure modes to reproducible SocketCAN error states and multimeter evidence in a published diagnostic catalog.

Research Publications

Sampling Budget Biases Apparent Metastable-State Counts in Molecular Simulation

- Used controlled synthetic landscapes and two independent alanine-dipeptide trajectories to show how sampling budget can inflate apparent state counts; under review at JCTC, with a ChemRxiv preprint and Zenodo dataset.

One Byte, One Rank Reversal: Auditing HumanEval Pipeline Sensitivity

- In a pinned five-model RTX 4070 reproduction, StarCoder2-3B scored 3/164 against its published 31.7%; removing one trailing prompt newline raised it to 49/164 and restored the ranking.

When Does INT8 Actually Accelerate Host-CPU Inference?

- Audited MobileNetV2 INT8 conversion and packaging, showing that INT8 latency depends on backend delegation and quantization granularity; v1.2.0 archived on Zenodo.

Conditioning and Directionality in Adversarial Transfer on CIFAR-10

- Evaluated directional transfer across three CNN families with multiple seeds and denominator-aware metrics; revising the manuscript following a TMLR rejection.

Not All Bad Demonstrations Are Equally Bad

- Ran 660 controlled policy fits across five demonstration failure modes, two policy families, ten seeds, and transition-matched controls; under review at the CoRL 2026 LfC workshop.

Smoothness Ranks Skill, Not Success

- Audited published trajectory-smoothness metrics under episode-length controls and found they primarily rank operator skill rather than task success; submitted in venue-specific forms to three workshops.

Technical Skills

Robotics & Hardware Integration: CAN/SocketCAN, RTDE, URSim, teleoperation, real-time control, fault injection, actuator control, telemetry analysis, electrical/mechanical troubleshooting, PCB rework

Machine Learning & Physical AI: PyTorch, JAX, LeRobot, OpenPI, LoRA, NumPy, pandas, scikit-learn, ONNX, model evaluation

Software, Systems & Data: Linux, Git, Docker, ROS2, Kubernetes, Redis Streams, PostgreSQL, REST APIs, AWS

Languages: Python, C++, TypeScript, JavaScript, SQL, Bash

Education

Grand Canyon University

M.S. in Computer Software Engineering

May 2024 – Oct 2025

Phoenix, AZ

- Capstone: Developed an AI model-packaging CLI that converts PyTorch, TensorFlow, ONNX, and scikit-learn models into deployable Docker images or Python packages.

University of Silicon Valley

B.S. in Computer Science

May 2021 – Aug 2023

San Jose, CA

- Capstone: Developed a MetaHuman lip-sync and audio-animation pipeline in Unreal Engine for real-time playback.

Certifications

Accelerated Computing in Modern CUDA C++ (Sep 2026)

Cisco Network Support and Security (Sep 2025)

Stanford AI and Machine Learning in Healthcare (Jun 2025)

Google IT Support Professional Certificate (May 2021)

HackerRank Software Engineer Certificate (Mar 2026)

IBM Artificial Intelligence Fundamentals (Aug 2025)

Google Foundations of Project Management (Sep 2024)